

Homework — 1.3.2 Databases

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.3.2 lesson. Show your working.

Part A — This week's topic (~14 marks)

A sports club uses the following two tables. Primary keys are shown in bold.

Member(**MemberID** , FirstName, Surname, MembershipType) Booking(**BookingID** , MemberID, CourtNumber, BookingDate)

1. Write the letter of the correct purpose in the **Match** column next to each SQL keyword. Each letter is used exactly once. [3] (AO1)

SQL keyword	Match	Letter	Purpose
SELECT		A	Adds a new record to a table
FROM		B	Filters the rows so only those meeting a condition are returned
WHERE		C	Combines rows from two tables using a linking condition between them
ORDER BY		D	States which fields (attributes) the query returns
JOIN ... ON		E	Sorts the results by a named field
INSERT INTO		F	Specifies the table the data is read from

2. The query below should return the **FirstName** and **Surname** of every member whose **MembershipType** is 'Junior', sorted by **Surname** in **descending** order. Fill in the blanks (a)–(d). [2] (AO2)

```
SELECT FirstName, (a)
FROM (b)
WHERE (c) = 'Junior'
ORDER BY Surname (d);
```

3. The query below is meant to list each member's **Surname** together with the **CourtNumber** and **BookingDate** of every booking they have made. It contains **three** errors. Identify each error and write the correction. [3] (AO2)

```
SELECT Member.Surname, Booking.CourtNumber, Booking.BookingDate
FROM Member
JOIN Booking WHERE Member.MemberID = Booking.BookingID;
```

.....

.....

.....

4. A mail-order company records each order as a single record, where one order can contain several products:

```
Order(OrderID, CustomerName, CustomerAddress, ProductID, ProductName, Quantity)
```

(a) Explain why this table is **not** in First Normal Form (1NF), and state what 1NF requires. [2] (AO1/AO2)

(b) After splitting into order lines keyed on (OrderID, ProductID), the attribute ProductName depends on ProductID alone — only part of the key. Name this type of dependency and state which normal form removes it. [2] (AO1)

5. The club's booking system processes payments as **transactions**. Explain the **Durability** property of ACID and why it matters if the server loses power. [2] (AO1/AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. (1.3.1) A digital photograph is to be uploaded to a website. State whether **lossy** or **lossless** compression is the more suitable choice and give **one** reason. [2] (AO1/AO2)

7. (1.2.3) State what is meant by the **Waterfall** model of software development and give **one** drawback of it. [2] (AO1)

8. (1.1.3) Explain **one** reason why an SSD might be chosen over a magnetic hard disk drive in a laptop.
[2] (AO1)
